

IMC 2025 Shortlisted Problems

- Two different normals of the parabola

$$2y = ax^2 (a > 0),$$

at the point B and B' of the parabola intersects the parabola again at a new point A with ordinate m . Prove that $m > \frac{4}{a}$.

- See problem 1 from IMC 2025;

- Let ω be a fixed circle on the plane, $\alpha \in (0, \pi/2)$ be a fixed acute angle. Ivan prefers isosceles triangles with angles $\alpha, \alpha, \pi - 2\alpha$, inscribed to ω . Ivan wants to mark several points on the plane so that every preferred triangle contains (inside or on the boundary) at least one marked point. Find the least number of marked points for which he can achieve this.

- Given an n -simplex with vertices $A_1, A_2, \dots, A_{n+1}$ ($n \geq 2$) and any point M in its interior, let N_i be the points on the hyperplane opposite A_i such that A_i, M , and N_i are collinear. Find the point M for which

$$\frac{\text{Volume}(MN_1N_2 \dots N_n)}{\text{Volume}(A_1A_2 \dots A_{n+1})}$$

is maximal and find this maximum.

- Let $\mathcal{R}, \mathcal{G}, \mathcal{B}$ be three finite, non-empty families of lines in the Euclidean plane, coloured red, green and blue respectively. A unit circle hits a line if the circle meets the line in at least one point. For every colourful triple $(\ell_r, \ell_g, \ell_b) \in \mathcal{R} \times \mathcal{G} \times \mathcal{B}$ there exists a unit circle that hits the three lines (ℓ_r, ℓ_g, ℓ_b) simultaneously. Prove that at least one of the three considered classes $\mathcal{R}, \mathcal{G}, \mathcal{B}$ can be hit in its entirety by a single unit circle.

- Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for every $x, y \in \mathbb{R}$,

$$(x - y)f(x + y) = (x + y)(f(x) - f(y)).$$

- Find all polynomials $f(x) = ax^2 + bx + c$ with $a, b, c \in \mathbb{R}$ that satisfy the following condition: for every real numbers x, y, z such that $x + y + z = xy + yz + zx > 0$ then we always have

$$f(x) + f(y) + f(z) \geq 2f(1) + f(xyz).$$

- Let $x_0 \geq 0$ and define the sequence $(x_n)_{n \geq 0}$ recursively by

$$x_{n+1} = \frac{x_n}{1 + \sqrt{1 + x_n^2}} \quad (n = 0, 1, 2, \dots).$$

(i) Prove that $x_n \rightarrow 0$ as $n \rightarrow +\infty$;

(ii) Evaluate the $\lim_{n \rightarrow +\infty} 2^n x_n$ in terms of x_0 .

- Let a be a positive integer number. Prove that $x_1^2 + \dots + x_n^2 - ax_1 \dots x_n \leq 0$ for $2 \leq x_1 \leq \dots \leq x_n \leq \frac{ax_1 \dots x_n}{2}$, $n \geq 4$. Find the cases of equality.

10. Find all strictly monotonic functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x)f^{-1}(x) = x^2.$$

11. Find all differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$, such that for all $x, y \in \mathbb{R}$ the inequality $f(x+y) \geq 3^x f(y) + 2^y f(x)$ holds true and $f(0) = 0$, $f(1) = 1$.

12. See problem 2 from IMC 2025;

13. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a differentiable function satisfying:

- 1) $f(0) = 0$ and $f(1) = 1$,
- 2) $f'(x) > 0$ for all $x \in (0, 1)$.

Prove that for every positive integer n , there exist distinct points $x_1, x_2, \dots, x_n \in (0, 1)$ such that

$$\sum_{k=1}^n \frac{1}{f'(x_k)} = n.$$

14. Consider the differential equation $dY/dx = Y^2$ for $n \times n$ matrices. Does there exist a non-degenerate (i.e. with $\det(Y) \neq 0$) solution defined on the whole real axis? Give the answers (i) for $n = 2024$ and (ii) for $n = 2025$.

15. Let $f(x) \in C^2(0; +\infty)$ and $f(x)$ has a positive second derivative. Prove that for every $t \in (0, \frac{1}{2})$ following inequality is satisfied

$$\int_0^{1-t} f(s)ds \leq (1+t-2t^2) \int_0^{1/2} f(s)ds + (1-3t+2t^2) \int_{1/2}^1 f(s)ds.$$

16. Let $g_1 : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that $\int_0^{+\infty} |g_1(t)|dt < +\infty$ and $g_1(0) = 1$. For every integer $n \geq 2$ define

$$g_n(x) = \int_0^x (x-t)g_{n-1}(t)dt, \quad x \geq 0.$$

Show that the $\lim_{n \rightarrow \infty} ng_n(1)$ exists and compute its value.

17. Let $\alpha_1, \alpha_2, \dots, \alpha_n$ be the non-zero distinct roots of the polynomial $p(x)$ of the degree of n with complex coefficients. Prove that the following equality holds

$$1 = \frac{p(1)}{p(0)} + \sum_{i=1}^n \frac{p(\alpha_i + 1)}{\alpha_i \cdot p'(\alpha_i)}.$$

18. (?) Find the smallest positive degree of a polynomial of the form

$$f(x) = (1+x)(1+x^2)(1+x^3)(1+x^4)$$

for which there exists $A \in \mathbb{Z}$ and polynomials $g(x), h(x) \in \mathbb{Z}[x]$ such that

$$f(x) = A + (x^2 - 1)g(x) + 20h(x)$$

19. An ordered triple (A, B, C) of positive integers is preferred if there exist the complex polynomials $a(x)$, $b(x)$, $c(x)$ with degrees A, B, C respectively and non-complex polynomials $p(x), q(x), r(x)$ for which

$$p + aq + br = ap + q + cr = bp + cq + r = 0.$$

Find the number of preferred triples with $A + B + C \leq 2025$.

20. Determine all positive integers n for which there exists a polynomial $P(x)$ with integer coefficients of degree at least 100, such that for each $1 \geq k \geq n$, the polynomial $(x + k)(P(x) + k)$ is a perfect square in $\mathbb{Z}[x]$.
21. Let $a_1, a_2, \dots, a_n$ be a non-constant arithmetic progression of real numbers. Prove that the determinant of an arbitrary $n \times n$ matrix composed line by line from different cyclic permutation of numbers $a_1, a_2, \dots, a_n$ is zero if and only if $a_1 + a_n = 0$.
22. See problem 8 from IMC 2025.
23. Let A, B be $n \times n$ real matrices. Prove that if all the eigenvalues of $AB - BA$ are real and

$$\text{tr}((AB)^2) = \text{tr}(A^2B^2),$$

then $AB - BA$ is nilpotent.

24. Let B be $n \times n$ positive-definite matrix with real entries. It turned out that

$$BC^2 - B^2C = E$$

and

$$\text{tr}(C^2) - \text{tr}(CB) \cdot \text{tr}(B) = 2025.$$

Find maximum possible value of n .

25. (?) Let $A, B \in M_n(\mathbb{R})$ satisfy $A^2 = E$, $B^2 = E$, $AB = BA$, $A \neq \pm E$, $B \neq \pm E$ and ... Prove that $|\text{tr}(C)| \leq n$.
26. Let A be a square matrix over a field F , $A^n = 0$. Prove that the equation

$$\sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \alpha_{ij} A^i X A^j = B, \text{ where } \alpha_{ij} \in F, \alpha_{00} \neq 0,$$

is solvable for any square matrix B of the same order as matrix A .

27. (?) Consider $\mathcal{B}$ the set of all real, symmetric matrices $B \in \mathcal{M}_{2025}(\pm 1)$ such that

$$\det(A + B) \cdot \det(A - B) \leq \det(A^2), \quad \forall A \in \mathcal{M}_n(?).$$

- a) Prove that if $B \in \mathcal{B}$, then $\text{rank}(B) = 1$.
- b) Determine the number of matrices $B \in \mathcal{B}$ with the property...
28. a) Prove there exists no real invertible $n \times n$ matrix A such that $A^T + A^{-1} = A$.

- b) Find all positive integers n such that there exists an invertible $n \times n$ real matrix A such that $A + A^{-1} = A^T$.
29. Find three consecutive primes which are all quadratic residues of each other.
30. Let $d(n)$ be the number of "2" in the prime factorization of $(n-1)!(n-2)! \dots 2111$. Find $d(32^{2025})$.
31. Let R be a unital ring, and let $n \geq 2$ be an integer. Let $x \in R$ have the property that $x^n = x$. Prove that for any $m \geq 1$, there exists a unit $u_m \in R^*$ such that $x^m = u_m x$.

32. Let $\pi : G \rightarrow Q$ be a surjective homomorphism of finite groups. Prove that there exists a subgroup H of G such that $\pi(H) = Q$, and the set of primes dividing $O(H)$ is the same as the set of primes dividing $O(Q)$.

33. Let $P(x) = x^2 + x$. Let S_N be the number of integers $1 \leq a, b \leq N$ such that $P(a)P(b)$ is a perfect square. Prove that the limit

$$\ell = \lim_{N \rightarrow \infty} \frac{S_N}{N}$$

exists and find its value. Is it true that for all large enough positive integers N , we have

$$|S_N - \ell N| \leq N^{3/2}?$$

34. Let G be the free product $\mathbb{Z}_3 * \mathbb{Z}_3$. If N_n denotes the number of subgroups of index n in G , prove that N_n is odd if, and only if, n is of the form $2^k - 3$.
35. Does there exist a rational number x such that

$$Q(x) = 2x^6 - x^4 + 3x^3 - 4x^2 - 6x - 4$$

is the square of a rational number?

36. Let p be a prime number. The polynomial $f(x) \in \mathbb{Z}[x]$ is called k -full, if $p^k \mid f(x)$ for any $x \in \mathbb{Z}$. Find the largest value of k for which the following implication is valid:

$$f \text{ is } k\text{-full} \quad \Rightarrow \quad f' \text{ is } (k-1)\text{-full, where } f' \text{ is the derivative of } f.$$

37. Let p be an odd prime number, and $q > 1$ be a divisor of $p-1$. Denote by T_q the number of residues $a \in \{1, \dots, p-1\}$ modulo p such that the order of the multiplicative group $\langle a \rangle$ generated by a is divided by q . Assume that

$$q \cdot T_q = (q-1)(p-1).$$

Show that q is a prime number if and only if q^2 does not divide $p-1$.

38. A sequence $a_1, a_2, \dots, a_n$ is called bright, if all its members are natural and the numbers

$$a_1 + 1, a_2 + 2, \dots, a_n + n$$

give all possible remainders from division by n . Let the sequence $a_1, a_2, \dots, a_{2025}$ be bright and $a_1 = 1, a_2 = 2, \dots, a_9 = 9$. What minimal value could have the sum $a_1 + a_2 + \dots + a_{2025}$?

39. Let S be the set of integer points on the coordinate plane Π (so $S \subseteq \mathbb{Z}^2$) and n be some positive integer. It turned out that for every $m > n$ one can find m vertical and m horizontal lines on Π such that if M is the set of integer points formed by the intersection of these lines, then $|M \cap S| > m^{1.01}$. Prove that one can find $m > n$ and m vertical and m horizontal lines on Π such that if N is the set of integer points formed the intersection of these lines, then $N \subseteq S$.
40. Let n, k be a positive integers. Alice and Bob are playing a game with n jars and k coins
- At the start of the game, for each coin Alice chooses which jar to place it in and whether to place it heads up or tails up.
 - Then, Bob may perform the following move as many times as he wishes: he picks any coin from any jar and moves it to any other jar of his choice. However, when the coin is placed in the new jar, Alice chooses whether it lands heads up or tails up.
 - Bob wins if he can reach a state where, for each jar, all the coins inside it have the same state (i.e. for each jar, all its coins are heads up or all are tails up).

Find all n, k for which Bob can win.

41. Suppose $A_1, A_2, A_3, \dots$ is a sequence of intervals contained in $[0, 1[$ of lengths $a_1, a_2, a_3, \dots$ respectively. Suppose that

$$\sum_{n=1}^{+\infty} \frac{a_n}{n} < +\infty.$$

Prove that the complement of the set of $\alpha \in [0, 1]$ for which

$$\lim_{N \rightarrow +\infty} \frac{\#\{n < N \mid \alpha \in A_n\}}{N} = 0,$$

is a zero set. (A set $S \subset [0, 1]$ is a zero set if for any $\varepsilon > 0$ there exists a collection of intervals $I_1, I_2, I_3, \dots$ of lengths $l_1, l_2, l_3, \dots$ respectively, such that $S \subseteq \bigcup_{n=1}^{+\infty} I_n$ and $\sum_{n=1}^{+\infty} l_n < \varepsilon$).

42. Let n be a positive integer, $M = \{1, 2, \dots, n\}$. A family $\mathcal{F}$ of subsets of M separates points if for every two distinct elements $a, b \in M$ there exists a set $A \in \mathcal{F}$ which contains exactly one of them. Denote by

$$\mathcal{F}^* = \{A \Delta B : A, B \in \mathcal{F}, A \neq B\}$$

the family of mutual symmetric differences of distinct sets from $\mathcal{F}$. Find the minimal possible size of $\mathcal{F}^*$, if $\mathcal{F}$ separates points.

43. Let G be a connected, simple graph on n vertices. For every vertex $v \in V(G)$, with d_v we denote its degree, that is, the number of edges connected to it. If G is not a regular graph, then prove that

$$\sum_{\{u,v\} \subseteq V(G)} (d_u - d_v)^2 \geq \begin{cases} n-1, & \text{if } n \text{ is odd;} \\ 2n-4, & \text{if } n \text{ is even.} \end{cases}$$

44. See problem 5 from IMC 2025.

45. A positive integer $n > 1$ is called “good” if the set $\{1, 2, 3, \dots, 3n\}$ can be divided into ordered triples, denote by (a_i, b_i, c_i) , $1 \leq i \leq n$, such that all n quadratic equations

$$a_i x^2 + b_i x + c_i = 0$$

have common roots. Prove that there exist infinitely many positive integers n that are good and also infinitely many positive integers that are not good.

46. For a graph $G = (V, E)$, we define

$$\sigma(G) = \sum_{uv \in E(G)} (d(u) - d(v))^2 \quad \text{and} \quad \sigma_t(G) = \sum_{\{u,v\} \subseteq V(G)} (d(u) - d(v))^2.$$

Find the smallest $\alpha > 0$ such that there is some $C > 0$ such that

$$\sigma_t(G) < Cn^\alpha \sigma(G)$$

for every graph G of order n .

47. Let k and n be positive integers. A teacher given to a student $2k$ (not necessarily distinct) numbers $a_1, \dots, a_{2k}$ which are known to be equal to $\cos x_1, \cos x_2, \dots, \cos x_k, \sin x_1, \sin x_2, \dots, \sin x_k$ in some order for certain real $x_1, \dots, x_k$. The student’s goal is to find $\cos nx_1, \cos nx_2, \dots, \cos nx_k, \sin nx_1, \sin nx_2, \dots, \sin nx_k$ in some order. Find all n such that the student can achieve the goal for every k and very real $x_1, \dots, x_k$.

48. Let G be a graph on n vertices and e edges. For a vertex $v \in V(G)$, let d_v denote its degree. Prove that following inequality

$$e \left(e + C_{n-1}^2 - (n-1) \right) \geq (n-1) \sum_{v \in V(G)} C_{d_v}^2.$$

Find all graph for which equality is attained.

49. Prove that for every two independent and identically distributed real random variables X and Y which take values in $\mathbb{Z}$,

$$\mathbb{P}(|X - Y| \leq 2) \leq 3\mathbb{P}(|X - Y| \leq 1).$$

50. A billiard table is convex and has a smooth boundary. Prove that there is an infinite number of different closed trajectories of the billiard ball (in other words. an infinite number of closed broken lines inside the table such that every vertex lies on the boundary, and the two sides meeting at that vertex are symmetric with respect to the outward normal).

51. Let $ABCD$ be a convex quadrilateral of total area K and let P be an interior point. Denote

$$A_1 = [PAB], \quad A_2 = [PBC], \quad A_3 = [PCD], \quad A_4 = [PDA],$$

where oriented areas are taken counter-clockwise so that all four A_i have the same sign. Show that

- (i) there exists exactly one interior point P for which the four areas A_1, A_2, A_3, A_4 form an arithmetic progression, and
- (ii) for this point the common difference is

$$d = A_2 - A_1 = A_3 - A_2 = A_4 - A_3 = \frac{[ACD] - [ABC]}{4} = \frac{K}{4}(1 - 2\mu), \quad \mu := \frac{[ABC]}{K}.$$

52. A natural number is called *Lego* if it can be represented as a sum of cubes of the first natural numbers. For example, 2025 is Lego since

$$2025 = 1^3 + 2^3 + \cdots + 9^3.$$

Find the sum of the reciprocals of all Lego numbers.

53. Let $n \geq 1$ be an integer and suppose some sequence of reals $a_1, \dots, a_n$ is given. Now define a_k for any $k > n$ as the mean of the n preceding terms in the sequence. Show that the sequence (a_k) converges and determine its limit.

54. See problem 7 from IMC 2025;

55. See problem 4 from IMC 2025;

56. • Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is a bijective function so that $f(ax + f(x))$ is continuous for any $a > 0$. Does it follow that f itself is continuous?
- Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function so that $f(ax + f(x))$ is bijective and continuous for any $a > 0$. Does it follow that f itself is bijective and continuous?

57. Let $a_1, a_2, \dots, a_n > 0$ be a positive real numbers with $a_1 + \cdots + a_n = 1$ and let $S_k = \sum_{j=1}^k a_j$ for $1 \leq k \leq n$. Prove that

$$\sum_{k=1}^n \frac{a_k}{1 + (S_k)^2} < \frac{\pi}{4}.$$

58. Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous function. Define $f_0(x) = x$ and $f_n(x) = f(f_{n-1}(x))$ for an integer $n \geq 1$. Find all pairs (k, a) , where k is a positive integer and $a \in [0, 1]$ such that there exists such a function f that

$$f_0(x) + f_1(x) + \cdots + f_{k-1}(x) = k \cdot a$$

for all $x \in [0, 1]$.

59. Can a continuous functions $f : \mathbb{S} \rightarrow \mathbb{R}$, where $\mathbb{S}$ is the circle, take all of its values the same (finite) number of times?

60. Let $f : [0, 1]^3 \rightarrow \mathbb{R}$ be a continuous function and

$$\iiint_{[0,1]^2} f(x, y, z) dx dy dz = \iint_{[0,1]^2} f(x, y, z) \cdot dx dy dz = \iint_{[0,1]^2} f(x, y, z) dz dx = 0,$$

for all $x, y, z \in [0, 1]$. Prove that

$$\iiint_{[0,1]^3} f^3(x, y, z) dx dy dz \geq \iint_{[0,1]^2} dy dz \left(\int_0^1 f(x, y, z) dy \right)^2 +$$

$$+ \iint_{[0,1]^2} dz dx \left(\int_0^1 f(x, y, z) dy \right)^2 + \iint_{[0,1]^2} f dx dy \left(\int_0^1 f(x, y, z) dz \right)^2.$$

61. Let $P(z) = z^2 + z$. For $n \in \mathbb{N}$ we define the set

$$S_n := \{z \in \mathbb{C} \mid \underbrace{P(P(\cdots P(z) \cdots))}_{n \text{ times}} = 0\}.$$

Determine the sum of all complex numbers in S_{2025} .

62. See problem 2 from VJIMC 2026.

63. Let n be a positive integer. Find the least possible positive integer k satisfying the following property: For every $n \times n$ complex matrix A , there is at least one invertible matrix among the k matrices

$$A, \quad A + I_n, \quad A + 2I_n, \quad \dots, \quad A + (k-1)I_n.$$

(I_n denotes the $n \times n$ identity matrix.)

64. Let D_n be the determinant of the $n \times n$ matrix A_n where the diagonal entries are all 3, the entries on the superdiagonal and subdiagonal are all 1, and all other entries are 0. Find a closed-form expression for $D_n = \det(A_n)$. For example:

$$A_4 = \begin{pmatrix} 3 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 1 & 3 \end{pmatrix}.$$

65. Find the determinant of the $n \times n$ matrix whose ij -entry is equal to the sum of natural numbers n exceeding $\min(i, j)$ and coprime with $(i+1)(j+1)$.

66. Let $B : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ be a bilinear form with matrix $A = [B]_{\mathcal{E}}$ in the standard basis $\mathcal{E}$ of $\mathbb{R}^n$. Assume that for every invertible matrix $P \in GL_n(\mathbb{R})$ the matrices A and $A_F = P^T A P$ have the same minimal polynomial. Determine all such bilinear forms B .

67. Let n be a fixed positive integer and, for an integer parameter k , form the $n \times n$ board whose entry in row i and column j is

$$A_{ij} = i + j - k, \quad 1 \leq i, j \leq n.$$

A perfect selection is a set of n squares – one from every row and every column – whose entries are precisely the integers $\{1, 2, \dots, n\}$.

(I) Determine the set

$$K(n) = \{k \in \mathbb{Z} : \text{the board } (A_{ij})_{1 \leq i, j \leq n} \text{ admits a perfect selection}\}.$$

(II) For every n with $K(n) \neq \emptyset$ give an explicit perfect selection.

68. Let V be a finite-dimensional complex vector space and let $T \in L(V)$ be a linear operator satisfying the following property: For every T -invariant subspace $W \subseteq V$ with $W \neq \{0\}$, there exists a non-zero vector $w \in W$ such that $Tw = 0$.

Prove that T is nilpotent, i.e., that there exists $n \geq 1$ such that $T^n = 0$.

69. Let V be an n -dimensional vector space over $\mathbb{C}$, and let $\mathcal{A}$ be a set of commuting matrices on V such that:

- (1) Each $T \in \mathcal{A}$ is cyclic (i.e., there exists $v \in V$ with $\{v, Tv, \dots, T^{n-1}v\}$ forming a basis for V).
- (2) For any two distinct $T_1, T_2 \in \mathcal{A}$, their eigenspaces are transversal:

$$\ker(T_1 - \lambda I) \cap \ker(T_2 - \mu I) = \{0\} \quad \forall \lambda, \mu \in \mathbb{C}.$$

Prove that $\dim \text{span}(\mathcal{A}) \leq n$, and characterize when equality holds.

70. Let $P, Q, R \in O(3)$ be real orthogonal 3×3 matrices, i.e.

$$P^T P = Q^T Q = R^T R = I_3.$$

Show that the matrix equation

$$P + Q = R$$

has no solutions in $O(3)$.

71. See problem 3 from IMC 2025.

72. Let A, B be $n \times n$ matrices so that

$$A^n B - B A^n = A^m B - B A^m$$

for every $m, n \in \mathbb{N}$. Prove that A and B commute given that either A or $A - I$ is invertible.

73. Find all monic polynomials P with integer coefficients with the property that

$$P(d(n)) \mid d(P(n))$$

for all positive integers n such that $P(n) > 0$. Here, if m is a positive integer, we denote by $d(m)$ the number of its positive divisors.

74. Let p be a prime, and consider the group G consisting of 2×2 matrices

$$\begin{pmatrix} y^m & y^n f(y) \\ 0 & 1 \end{pmatrix}$$

where f varies in $\mathbb{F}_p[y]$ and m, n over integers.

- (i) Show that G is finitely generated;
- (ii) Show that there does not exist a finite sequence of elements $g_1, \dots, g_n$ (not necessarily distinct) of G such that

$$G = \{g_1^{k_1} g_2^{k_2} \cdots g_n^{k_n} : k_i \in \mathbb{Z}\}.$$

75. Let $f : \mathbb{N} \rightarrow \mathbb{Z}$ be a nonconstant function such that for each finite group G and any elements $g, h \in G$

$$f(o(g^{o(h)} \cdot h^{f(o(g))})) = f(o(h^{o(g)} \cdot g^{f(o(h))}))$$

holds, where $o(a)$ denotes the multiplicative order of an element a . Prove that $f(1)$ is odd.

76. Let $P(x) = x^2 + x$. Let S_N be the number of integers $1 \leq a, b \leq N$ such that $P(a)P(b)$ is a perfect square. Prove that the limit

$$\ell = \lim_{N \rightarrow \infty} \frac{S_N}{N}$$

exists and find its value. Is it true that for all large enough positive integers N , we have

$$|S_N - \ell N| \leq N^{\frac{1}{2025}} \quad \text{for large } N?$$

77. Consider functions

$$B_i(x) = \frac{p_i x + k_i}{q},$$

where $n > 1$, k_i are integers, p_i, q are non-zero integers such that

$$(p_i, q) = 1 \quad \text{for } i = 0, 1, \dots, n-1.$$

Consider equations $B_i \circ B_{i+1} \circ \dots \circ B_{i+n-1}(x_i) = x_i$, and their solutions x_i for $i = 0, 1, \dots, n-1$, where all the indices are taken modulo n .

1. Prove that all x_i for $i = 0, 1, \dots, n-1$ are rational numbers, and in their simplest forms, all of x_i are like fractions with the same denominator.
2. Prove that for every non-negative integer b there is a non-negative integer β , such that for any i , the number

$$x_i - q^b p_i \cdot p_{i+1} \cdot \dots \cdot p_{i+b-1} x_{i+b}$$

is an integer, where again all the indices are taken modulo n . Note that β and b do not depend on i .

78. Consider the permutations of the numbers $1, 2, \dots, n$. We identify all permutations which differ only by the symbols appearing after the symbol n . So, for $n = 3$, the permutations 312 and 321 are identified, but 132 and 231 are not. A *cut permutation* is a class defined by the chosen equivalence relation of identification. Let $R(n)$ be the number of cut permutations of $1, 2, \dots, n$ and compute

$$\frac{R(2024) - 1}{R(2023)}.$$

79. Let $A_1, A_2, \dots, A_{2025}$ be a convex polygon. Let B be such a set of points, that for any i, j, k ($1 \leq i < j < k \leq 2025$) there is at least one point which belongs to B and lies strictly inside the triangle $A_i A_j A_k$. What is the minimal possible number of points in B ?

80. See problem 9 from IMC 2025.

81. (i) Suppose that a convex polygon $A_0A_1A_2\ldots A_{n-1}$ ($n \geq 3$) is given. Choose an internal point on each side of this polygon. Denote these points by $B_i \in A_iA_{i+1}$, where B_i is between A_i and A_{i+1} , and $B_i \neq A_i, A_{i+1}$ ($i = 0, 1, \ldots, n-1$). For such polygons we say that $B_0B_1B_2\ldots B_{n-1}$ is inscribed into $A_0A_1A_2\ldots A_{n-1}$ and $A_0A_1A_2\ldots A_{n-1}$ is circumscribed around $B_0B_1B_2\ldots B_{n-1}$. In the same way choose on each side of polygon $B_0B_1B_2\ldots B_{n-1}$ a point $C_i \in B_iB_{i+1}$ (C_i is between B_i and B_{i+1} , and $C_i \neq B_i, B_{i+1}$ for $i = 0, 1, \ldots, n-1$) and draw polygon $C_0C_1C_2\ldots C_{n-1}$. Now fix polygons $A_0A_1A_2\ldots A_{n-1}$ and $C_0C_1C_2\ldots C_{n-1}$. Find the maximum of the number of convex polygons like $B_0B_1B_2\ldots B_{n-1}$, which are inscribed into $A_0A_1A_2\ldots A_{n-1}$ and circumscribed around $C_0C_1C_2\ldots C_{n-1}$ possibly with the indices shifted so that $C_i \in B_{k+i}B_{k+i+1}$ ($i = 0, 1, \ldots, n-1$) for some fixed k .

(ii) Solve the same problem for spherical convex polygons over a sphere.

82. 1. Show that if $0 \leq a \leq x < b$ and $n > 0$, then

$$\lim_{k \rightarrow +\infty} \frac{e^{kx^n}}{\int_a^b e^{kt^n} dt} = 0.$$

2. Show also that if $0 \leq a < x = b$, then the limit is equal to $+\infty$.

83. Let $x_1, y_1, z_1 \in \mathbb{R}$ and let $0 < \alpha < 1$. The sequences $\{x_n, y_n, z_n\}_{n=1}^{\infty}$ are given by the recurrence formulae:

$$x_{n+1} = \alpha x_n + (1 - \alpha)y_n, \quad y_{n+1} = \alpha y_n + (1 - \alpha)z_n, \quad z_{n+1} = \alpha z_n + (1 - \alpha)x_n \quad (n = 1, 2, \ldots).$$

Prove that the sequence $\{y_n\}_{n=1}^{\infty}$ converges and find its limit.

84. Let $f : [0, +\infty) \rightarrow [0, +\infty)$ be a continuous function such that $f(x) \rightarrow C$ ($C > 0$) as $x \rightarrow +\infty$. For every positive integer n , define the following numerical sequences:

$$\frac{1}{a_n} = \int_{n^2}^{n^2+1} (f(t))^n dt \quad \text{and} \quad \frac{1}{b_n} = \int_{(n-1)^2}^{(n-1)^2+1} (f(t))^n dt.$$

Find the limit

$$\lim_{n \rightarrow +\infty} \frac{b_1 + b_2 + \cdots + b_n}{a_1 + a_2 + \cdots + a_n}.$$

85. Let B and C be $n \times n$ positive semi-definite matrices with real entries. It turned out that

$$A(BC + CA) + (AC + B^2)C + (B + A)BA + BC(C + A) = I$$

and

$$(\text{tr } AC + \text{tr } BC + \text{tr } A^2 + \text{tr } AB)(\text{tr } B + \text{tr } C) = 2025.$$

Find the maximum possible value of n .

86. Let $\alpha \in \mathbb{R}$ satisfy $|\alpha| \leq 1$. Consider the sequence $a_0, a_1, \ldots$ defined by

$$a_0 = 1, \quad a_1 = \alpha, \quad a_{n+2} = 2\alpha \cdot a_{n+1} - a_n$$

for all integers $n \geq 0$. Prove that $|a_n| \leq 1$ for any integer $n \geq 0$.

87. 1) Let $f : (0, 1) \rightarrow \mathbb{R}$ be twice continuously differentiable. Prove that the following two statements are equivalent:
- (i) f is convex and $2|f'(x) - f'(y)| \leq |x - y|$ for all $x, y \in (0, 1)$;
 - (ii) $f(x) \geq f(y) + f'(y)(x - y) + |f'(x) - f'(y)|^2$ for all $x, y \in (0, 1)$.

88. Describe all the triplets (a, b, c) of positive integers greater than 1 having the following property: for any positive integers k and m there exists a positive integer l such that

$$\gcd(a^k - 1, b^m - 1) = c^l - 1.$$

89. Let a sequence of positive integers $\{a_n\}_{n=1}^{\infty}$ be defined by the initial values $a_1 = 1, a_2 = 1, a_3 = 2$ and by the recurrence relation for all $n \geq 4$:

$$a_n a_{n-2} a_{n-3} = a_{n-1}^2 a_{n-3} + a_{n-1} a_{n-2}^2.$$

Prove that the sum of the reciprocals of this sequence,

$$S = \sum_{n=1}^{\infty} \frac{1}{a_n},$$

is an irrational number.

90. Let $n \geq 2$ be a positive integer. For every ordered triple of positive integers (i, j, k) satisfying $i \leq j \leq k \leq n$, a label $c(i, j, k)$ is written, with $1 \leq c(i, j, k) \leq n$. Prove that there exists a label $m \in [n]$ that occurs on at least

$$\frac{(\log n)^2}{9}$$

distinct triples (i, j, k) with $i \leq j \leq k \leq n$. (Here $\log$ denotes the natural logarithm.)

91. Fix an integer $k \geq 1$. Let $\mathbb{F}$ be a field with characteristic $p \in \{0, 2, 3, \dots\}$ and let $n \geq 1$. Consider matrices

$$A \in M_n(\mathbb{F})$$

satisfying

$$\prod_{i=0}^k (A - iI_n) = 0_n. \quad (1)$$

- (i) Determine all characteristics p for which the following rank-sum identity is guaranteed to hold for every matrix A satisfying (1):

$$\sum_{i=0}^k \text{rank}(A - iI_n) = kn. \quad (2)$$

- (ii) For each characteristic in which (1) fails, construct an explicit counter-example A (for some n) that satisfies (1) but violates (2).

92. Fix the small constant $h = \frac{1}{2025}$. Let

$$C = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1, 0 \leq z \leq h\}$$

be the lateral surface of a right circular cylinder of radius 1 and height h . Every point of C is coloured with one of four colours. Prove that there exist two points on C of the same colour whose Euclidean distance is exactly 1.

93. Consider the unit cube $[0, 1]^3 \subset \mathbb{R}^3$. Four points are chosen independently and uniformly at random inside of this cube. What is the probability that the convex hull of these four points contains the center of the cube, $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$?

94. Fix a positive integer N and define

$$M_N = \{A = (a_{ij}) \in M_2(\mathbb{R}) : a_{ij} \in \{1, 2, \dots, N\}\}.$$

Two matrices $A, B \in M_N$ are chosen independently and uniformly at random.

(a) Show that the number of ordered pairs $(A, B) \in M_N^2$ that satisfy the commutativity condition $AB = BA$ is at most on the order of N^6 ; that is,

$$\#\{(A, B) \in M_N^2 : AB = BA\} = O(N^6) \quad (N \rightarrow \infty).$$

(b) Deduce that

$$\mathbb{P}(AB = BA) = O(N^{-2})$$

and hence

$$\lim_{N \rightarrow \infty} \mathbb{P}(AB = BA) = 0.$$

95. 1. Let ω denote the complex cube root of unity. Show that for z lying on the unit circle,

$$|z - 1|^2 |z - \omega|^2 + |z - 1|^2 + |z - \omega|^2$$

is a rational function of the form

$$\frac{\omega f(z)}{g(z)}.$$

2. Show that exactly two roots of $f(z)$ lie outside the unit disc.

3. If α_1, α_2 are the roots of $f(z)$ which lie outside the unit disc, prove that

$$(a) \quad \arg(\alpha_1 + \alpha_2) = \frac{\pi}{3};$$

$$(b) \quad |\alpha_1 \alpha_2| > 5;$$

$$(c) \quad |\alpha_1| = |\alpha_2|.$$

96. Let G be a finite group satisfying:

1. For every $g \in G$, $g^2 = e$ (all elements have order 2);

2. G acts faithfully on $\mathbb{R}^n$ via linear transformations (i.e., there exists an injective homomorphism $G \rightarrow \text{GL}_n(\mathbb{R})$).

Prove that:

(a) G is abelian;

(b) $|G| \leq 2^n$;

(c) Describe all such groups G when equality $|G| = 2^n$ holds.

97. See problem 6 from IMC 2025.

98. Compute the limit

$$\lim_{n \rightarrow \infty} \frac{n^1 + n^2 + \cdots + n^n}{1^n + 2^n + \cdots + n^n}.$$